



Challenges and Costs of Electrifying the Italian Vehicle Fleet

Luca Piancastelli*

Department of Industrial Engineering (DIN), Università di Bologna Alma Mater Studiorum, 40136 Bologna, Italy

* Correspondence: Luca Piancastelli (luca.piancastelli@unibo.it)

Received: 11-20-2025

Revised: 01-15-2026

Accepted: 01-29-2026

Citation: L. Piancastelli, “Challenges and costs of electrifying the Italian vehicle fleet,” *Power Eng. Eng. Thermophys.*, vol. 5, no. 1, pp. 18–33, 2026. <https://doi.org/10.56578/peet050102>.



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Abstract: This study provides a quantitative assessment of the technical and economic implications of converting the entire Italian vehicle fleet to full electric power. Investment estimates for night-time-only charging indicate a total requirement of approximately \$208.0 billion, including \$194.4 billion for generation capacity and \$13.6 billion for network reinforcement. For daytime-only fast charging at 280,000 MW, the total investment rises to approximately \$627.9 billion, with \$604.8 billion allocated to generation and \$23.1 billion to network upgrades. The combined total for both scenarios reaches approximately \$835.9 billion, underscoring the dominant role of generation in the overall expenditure. The analysis highlights that even under conservative assumptions, the expansion of installed power capacity and the doubling of supply points required for nighttime charging, along with peak power requirements up to five times current grid capabilities for daytime charging, exceed realistic infrastructure limits. The economic burden of such investments would largely fall on taxpayers and may be incompatible with the national economy. The study further suggests that the accelerated adoption of fully electric vehicles, without considering broader grid constraints and operational limits, may produce secondary effects more severe than the intended environmental benefits. A diversified strategy, incorporating hybrid systems, synthetic fuels, hydrogen, or improved internal combustion technologies, is recommended to mitigate infrastructure pressure and reduce economic risks. The work is presented as a conservative initial assessment, intended to stimulate further research on energy, infrastructure, and economic impacts to support technically feasible and economically sustainable transition strategies for the national automotive system.

Keywords: Electric vehicle charging demand; Peak power requirements; Fast-charging infrastructure; Energy demand analysis; Grid capacity assessment; Load management strategies; Power system planning

1 Introduction

This work is built upon previous studies. The first study [1] examined the transformation of a traditional automobile into a smartphone-like system equipped with a touch screen and a battery, showing that the use of touch-screen interfaces is incompatible with driving safety. The second study [2] verified and extended the analysis of the charging power and the charging energy required to support a fully electric vehicle fleet in Italy. This third study considers the costs of strengthening the national electrical grid in the event of a complete replacement of the current thermal vehicle fleet with a fully electric one. A key result of the second study is that the national electrical grid would need to be at least doubled in capacity to support exclusively nighttime charging for an entirely electric vehicle fleet. Moreover, the maximum available power would need to be increased by a factor of four if daytime charging were also to be enabled, especially during high-travel-demand periods, such as the so-called red-alert travel days. A simplified economic assessment of the national electrical grid is presented with reference to a required increase of at least 40% in available power and an approximate doubling of supply points, from about 30,000 to roughly 70,000. The estimate assumes a fleet of approximately 40 million vehicles in Italy and remains intentionally conservative, as the assumption of one charging point per vehicle is unlikely to be sufficient. All issues related to the physical installation, operation, security, and protection of charging stations against improper use fall outside the scope of this analysis. The purpose of the present work is to estimate, in an approximate manner, the cost of doubling the national grid both in terms of supply points and in terms of maximum deliverable power. A further estimate is given for the scenario in which daytime charging capability is also provided. Daytime charging requires the use of high-power devices, typically connected to medium- or high-voltage lines (15 kV and above), which are

extremely expensive. Their cost is not included here, neither for domestic users, whose minimum meter rating is 5 kW, nor for rapid charging installations, which require dedicated and costly equipment. The problem is particularly severe for rapid charging. Not only would the grid need four to five times the current maximum power, but rapid charging devices are expensive, and their throughput is low: while a single fuel pump can serve up to 20 vehicles per hour, a rapid charging point typically serves only one vehicle per hour. In theory, each fuel pump would need to be replaced by up to 20 rapid charging units. Commercial claims of 10-minute or 20-minute full battery charges refer to battery technologies that do not currently exist on the market. Faster charging results in lower efficiency and higher battery heating. To prevent overheating, the vehicle must actively cool the battery, increasing both power demand and energy losses. In practice, lithium batteries should not exceed about 40 °C during charging and ideally should remain within 30–40 °C. Extremely fast charging at rates such as 10C is therefore impractical: for example, charging a 100-kWh battery in six minutes would require about 1 MW of installed charging power, which is unrealistic for widespread infrastructure. Additionally, frequent rapid charging significantly reduces battery life. Thus, although many statements found in commercial advertising are theoretically correct, they do not reflect real operating conditions.

1.1 Organization of the Paper

This paper is structured as follows. A fully electric vehicle fleet is assumed. In the first part, the costs associated with reinforcing the Italian national power grid to support nighttime-only vehicle charging are analyzed. In the second part, the costs associated with reinforcing the Italian power grid due to daytime fast charging are examined. In both cases, the analysis provides an approximate cost assessment, which is divided into the costs for installing additional power generation capacity and the costs for implementing and upgrading the electrical transmission and distribution network.

2 Reinforcement of the Italian Power Grid for Nighttime-Only Charging

Nighttime charging would likely require a substantial increase in the number of users connected to the Italian power grid, which is currently designed to serve approximately 30 million users. Given that the Italian vehicle fleet comprises more than 40 million vehicles, a conservative estimate suggests that the number of grid connections may need to be approximately doubled, reaching about 60 million. This estimate does not account for the potential need for multiple charging points per vehicle, nor does it consider the additional reinforcement that would likely be required to accommodate a vehicle-oriented electrical network with higher power demands. Such a network would plausibly require electricity meters with a minimum contracted power of 5 kW, compared to the current standard value of approximately 2.5 kW.

2.1 Installed Power Capacity for Nighttime-Only Charging

The installed power capacity required to support nighttime vehicle charging is estimated to be approximately twice the current peak power capacity of the Italian electrical system. Unfortunately, as will be discussed in the following sections, it does not appear feasible in the Italian context to significantly increase the contribution of renewable energy sources for this purpose. Specifically, the large-scale expansion of wind power is limited by both economic factors and environmental impact associated with wind farm installations. Similarly, the deployment of photovoltaic (PV) systems is constrained primarily by economic considerations. Hydroelectric power faces substantial environmental and political constraints, while biomass-based generation has already reached a level that is close to the maximum compatible with the current Italian economic framework. As a result, the additional power demand would likely need to be met by fossil fuel-based generation, most plausibly natural gas, which represents the least polluting option among conventional fossil fuels. The economic estimates presented in this work, therefore, refer to this generation scenario.

2.2 Substantial Limitations to a Significant Increase in Renewable Power Generation in National Grids

The integration of large shares of renewable generation into national grids faces substantial technical, economic, and environmental constraints [3]. Variable and intermittent resources, such as wind and solar, pose challenges to grid stability, necessitating enhanced flexibility in both generation and demand management. While modern energy storage technologies, demand response programs, and advanced forecasting methods partially mitigate these challenges, the scalability of these solutions is often limited by cost, spatial availability, and technological maturity [4]. Grid infrastructure constitutes another critical limitation. Many national transmission and distribution networks were originally designed for centralized, dispatchable generation and may be insufficiently robust to accommodate high penetration of decentralized renewable resources. Upgrading or expanding grid capacity involves significant capital expenditure (CAPEX), regulatory approvals, and long construction lead times, which can impede rapid expansion [5]. In addition, congestion, voltage fluctuations, and reactive power management become increasingly significant as renewable penetration rises, necessitating advanced control systems and real-time monitoring. Economic and policy

factors further influence the pace and scale of renewable deployment. High upfront capital costs for generation and grid integration, exposure to volatile material and equipment prices, and uncertainties in electricity market prices can reduce the financial attractiveness of renewable projects [6]. Government incentives, feed-in tariffs, and long-term power purchase agreements often play a decisive role in supporting investment, particularly in nascent or capital-intensive technologies. Conversely, the gradual withdrawal of subsidies or fluctuations in policy frameworks may constrain investment and development. Environmental and social considerations also impose practical limits. The large-scale deployment of renewables can lead to land-use conflicts, ecological impacts, and community opposition, particularly for wind and hydroelectric projects. Site-specific factors such as solar irradiance, wind resource variability, water availability, and environmental sensitivity can determine the technical and economic feasibility of each project, creating inherent spatial constraints on deployment [7].

2.2.1 Economic viability of wind energy projects

While wind energy provides significant environmental and strategic advantages, its economic viability remains constrained by several structural and market-related factors [8]. Wind power projects are characterized by high upfront capital requirements, and their financial sustainability depends on complex interactions among technological, economic, and regulatory factors.

High initial expenditures for turbine procurement, transportation, installation, and transmission infrastructure must be offset by long-term electricity sales and stable policy frameworks. Government incentives, subsidies, and tax credits often play an important role in ensuring financial viability. However, project outcomes remain strongly influenced by site-specific wind conditions, regulatory stability, and electricity market dynamics. Performance degradation, unexpected component failures, and the need for backup generation to maintain grid reliability introduce additional operational and financial complexity.

Material price volatility has recently intensified these economic challenges. Increases in the prices of steel, copper, and critical minerals used in turbine construction have significantly raised manufacturing and installation costs. At the same time, higher global interest rates have increased the cost of capital, thereby reducing the attractiveness of capital-intensive wind investments. These pressures have been particularly evident in large offshore projects, where installation, logistics, and grid connection costs are considerably higher than in onshore developments.

Additional economic challenges arise from integrating intermittent wind generation into electrical power systems. Because wind power output varies with meteorological conditions, grid operators must maintain sufficient reserve generation capacity and invest in network upgrades to ensure system stability. These intermittency-driven costs, including grid reinforcement and balancing services, remain substantial in many regions.

Operational reliability and maintenance requirements represent another important factor affecting economic viability. Wind turbines are subjected to significant mechanical stress and environmental conditions throughout their operational lifetimes. Component failures, including gearbox and blade damage, may increase maintenance costs and reduce turbine availability. Although technological improvements and larger turbine designs can provide economies of scale, they also require greater initial investments and may introduce additional technical risks.

Recent developments in both the United States and Europe indicate that several major developers have incurred financial losses or withdrawn from planned wind energy projects due to rising costs and unfavorable financing conditions. These developments challenge earlier expectations of continuous cost reductions in the wind energy sector and highlight the ongoing need for robust policy mechanisms and resilient financial structures to ensure the sector's long-term sustainability.

2.2.2 Safety and environmental risks associated with wind turbines

Wind energy infrastructure, while contributing to renewable electricity generation, is not free from safety concerns related to accidents and fatalities. Reports compiled from documented incidents indicate that serious events have occurred across all phases of the wind turbine lifecycle, including construction, operation, maintenance, and transportation. A comprehensive dataset of turbine-related accidents reported up to September 2025 recorded thousands of incidents worldwide, a subset of which resulted in fatalities affecting both industry personnel and members of the public.

Analysis of available data suggests that accidents may result from mechanical failures, structural collapses, or transportation-related incidents during the installation or relocation of turbine components. Component failures, such as blade detachment or structural collapse, have been repeatedly documented and may pose hazardous conditions for workers and nearby communities. Independent news reports also document individual workplace fatalities, including incidents in which workers have died after falling from turbine structures during installation activities [9]. These events highlight the occupational risks associated with turbine construction and maintenance.

In addition to direct human injuries and fatalities, turbine operation has been associated with ecological impacts. Wildlife collisions with rotor blades have been documented in scientific studies focusing on avian and chiropteran mortality associated with wind energy installations [10]. These ecological effects are relevant to broader assessments of turbine safety and environmental sustainability.

Regulatory frameworks in several jurisdictions require the prompt reporting of accidents or technical failures that may pose safety risks. Such incidents include blade failures, turbine fires, or uncontrolled rotor motion, which must be reported to facilitate regulatory oversight and remedial action [11]. Nevertheless, disparities in reporting practices and the confidentiality of certain industry datasets have been identified as obstacles to achieving a comprehensive understanding of the frequency and severity of turbine-related accidents.

Addressing these challenges requires systematic risk assessment, improved transparency in accident reporting, and continuous improvements in turbine design, installation procedures, and maintenance practices. Such measures are essential to minimize both human and environmental risks associated with wind energy systems.

2.2.3 Solar economics

While solar energy offers significant environmental and strategic advantages, its economic viability is constrained by several structural and market factors [6]. High upfront capital requirements, volatility in material prices such as silicon, silver, and rare-earth elements used in inverters, increasing interest rates, and substantial grid-integration expenses have contributed to reduced profitability and, in several cases, to the cancellation or delay of planned projects. These pressures have been particularly evident in large utility-scale solar farms, where installation, logistics, and grid connection costs are higher.

Overall project viability depends on location, technology, policy conditions, and solar resource characteristics. High initial expenditures associated with photovoltaic module procurement, mounting structures, inverters, and transmission infrastructure must be offset by long-term electricity sales, incentives, and stable policy frameworks. However, performance degradation, inverter failures, and the requirement for storage or backup generation to ensure grid reliability introduce additional operational and financial complexity.

Inflation and material cost escalation have recently intensified these challenges. Increases in the prices of silicon, silver, and other critical minerals have significantly raised manufacturing and construction costs, compressing developers' margins. At the same time, higher interest rates have increased the cost of capital, reducing the attractiveness of large-scale solar investments. Intermittency-driven grid integration expenses, including network upgrades and reserve generation capacity, remain substantial for many regions.

Solar PV performance and maintenance concerns, such as inverter failures, soiling, or module degradation, further increase operational expenditures over the asset lifetime. Project outcomes are also shaped by site-specific solar irradiance, availability of government support through subsidies or tax credits, and advances in PV technology. Although larger and more efficient solar modules can achieve economies of scale, they also require greater initial investment and carry higher technical risk. Market conditions, including wholesale electricity prices and long-term power purchase agreements, play a decisive role in determining revenue stability.

Recent developments in both the United States and Europe have shown that several major developers are incurring financial losses or postponing planned installations due to these economic pressures. These trends challenge earlier expectations of continuous cost reductions and underscore the ongoing need for robust policy mechanisms and resilient financial structures to ensure the sector's long-term sustainability.

2.2.4 Solar safety and environmental impacts

Solar power infrastructure, while contributing to renewable electricity generation, is not free from safety concerns related to accidents and injuries. Reports indicate that incidents can occur during the lifecycle of solar installations, including construction, operation, maintenance, and transport. Documented accidents include falls during panel installation, electrical shocks, fires originating from faulty inverters or wiring, and vehicle accidents during the delivery and handling of solar modules [12]. These incidents highlight that, although solar power is widely considered a low-risk energy source, safety challenges remain and require systematic management.

Analysis of available data suggests that accidents can result from electrical faults, structural failures of mounting systems, and human error during maintenance or installation. Fires caused by inverter malfunctions or short circuits in photovoltaic arrays have been repeatedly reported, leading to property damage and potential hazards to workers [12]. Independent news reports also document fatalities and serious injuries during construction, including falls from elevated mounting structures and rooftop installations [9].

In addition to direct human risks, solar plants may also have ecological impacts, particularly if large-scale ground-mounted PV farms replace natural habitats, leading to changes in local biodiversity or soil compaction [7]. These environmental considerations are increasingly relevant to broader assessments of solar energy sustainability.

Regulatory frameworks across jurisdictions mandate reporting of safety incidents, including fires, electrical accidents, and structural failures, to facilitate oversight and remedial action [11]. Nevertheless, disparities in reporting practices and limited public access to incident data can hinder a comprehensive understanding of risk frequency and severity. Addressing these safety challenges requires rigorous risk assessment, enhanced reporting and transparency, and continual improvement of design, installation, and maintenance protocols to minimize both human and environmental harm associated with solar energy systems.

2.2.5 Hydropower environmental impacts

Although hydropower remains one of the most mature and widely deployed renewable energy technologies, its environmental impacts are the subject of extensive evaluation due to the ecological and geomorphological alterations it introduces across river basins. Large dams and associated reservoirs can significantly modify natural hydrological regimes, alter sediment transport, and transform aquatic and riparian habitats, generating complex environmental trade-offs that must be considered alongside hydropower's low operational carbon footprint [13]. Lifecycle analyses also indicate that reservoir formation may produce greenhouse gas emissions through the decomposition of flooded biomass, particularly in tropical or highly productive systems [14].

The environmental performance of hydropower installations depends strongly on dam scale, reservoir morphology, regional climate, and operational strategies. Hydrological regulation, including flow stabilization and peak-power releases, can disrupt downstream ecological processes by altering temperature regimes, dissolved oxygen levels, and seasonal flow patterns that are essential for fish migration and spawning. Furthermore, the impoundment of large volumes of water can lead to the loss of terrestrial habitats and the displacement of species, altering landscape connectivity and biodiversity [15]. These factors underscore the need for comprehensive basin-scale environmental assessments when planning new hydropower facilities or upgrading existing ones.

Sediment trapping is a persistent challenge in reservoir-based systems. By intercepting upstream sediment loads, dams can accelerate reservoir infilling, reduce storage capacity, and alter downstream fluvial geomorphology, potentially increasing erosion and degrading deltaic ecosystems [16]. Sediment management strategies such as flushing, bypassing, or dredging can mitigate impacts but often entail substantial operational and environmental costs. Water quality issues, including stratification and nutrient accumulation, may also arise in deep reservoirs, influencing downstream productivity and aquatic health.

Hydropower also poses risks to biodiversity. Migration barriers created by dams can drastically reduce populations of anadromous and catadromous fish species, while changes in flow variability can reduce habitat suitability for sensitive aquatic organisms [17]. Even small hydropower schemes may affect riverine ecosystems when deployed at high density. In addition, reservoir expansion can increase the likelihood of invasive species establishment by altering ecological conditions.

End-of-life management, operational retrofits, and dam decommissioning present further environmental considerations. As many hydropower facilities constructed in the mid-twentieth century approach structural or economic obsolescence, questions concerning sediment release, ecosystem restoration, and long-term stability have become central to environmental planning [18]. Ensuring that hydropower development remains environmentally sustainable requires integrated watershed governance, adaptive management, and continued technological improvements in turbine design, fish passage solutions, and sediment-handling techniques.

2.2.6 Hydropower safety and historical failures

Hydropower systems, despite their overall contribution to renewable electricity generation, are associated with safety and environmental risks that can arise during construction, operation, maintenance, and dam aging. Documented incidents indicate that structural failures, flooding events, turbine malfunctions, and spillway mismanagement can result in severe consequences for downstream communities and ecosystems. Historical analyses of dam-related accidents have shown that failures, though rare relative to the number of operational facilities, can lead to catastrophic loss of life and substantial environmental damage due to the sudden release of impounded water and sediment [19].

Mechanical or structural failures in hydropower plants, including turbine failures, generator fires, or pressure conduit ruptures, may also pose environmental hazards. Oil leaks from hydraulic systems, unintended releases of debris, and the mobilization of contaminated sediments can adversely affect water quality and aquatic habitats. Slope instability around reservoirs, occasionally triggered by rapid drawdown or prolonged saturation, can pose additional risks to infrastructure and surrounding environments [20]. These events illustrate the interconnected nature of structural integrity, operational safety, and environmental protection in hydropower systems.

Hydropower operations can further influence ecological conditions through rapid changes in flow associated with load-following or peaking power generation. Such hydropeaking events may cause fish stranding, alteration of benthic communities, and destabilization of riparian zones [21]. Reservoir-induced seismicity, though typically low-magnitude, has been documented in several regions and remains an area of active monitoring and research. Regulatory frameworks in many jurisdictions require dam operators to report structural anomalies, operational incidents, and environmental hazards, including turbine failures, emergency spillway use, and sudden changes in reservoir levels [11]. However, reporting practices and public access to safety data vary widely by country, complicating large-scale assessments of hydropower risk. Mitigating these risks necessitates rigorous structural monitoring, enhanced environmental oversight, transparent reporting, and ongoing investment in dam safety upgrades. Improvements in real-time hydrological forecasting, adaptive flow management, fish-friendly turbine technologies, and reservoir sediment strategies remain essential for aligning hydropower development with both environmental stewardship and

public safety.

Historical dam failures offer critical insights into the environmental and safety risks associated with large hydropower infrastructure. Among the most significant examples is the 1963 Vajont disaster in northern Italy, widely regarded as one of the most catastrophic failures in hydropower history. Although the dam itself did not collapse architecturally, the rapid displacement of a massive landslide into the reservoir generated a wave that overtopped the dam, resulting in the destruction of downstream settlements and causing nearly two thousand fatalities [22]. This event illustrates that hydropower risks may arise not only from structural deficiencies but also from complex interactions between geological instability, reservoir operations, and insufficient hazard assessment.

The Vajont case demonstrates how reservoir-induced changes in pore pressure, combined with pre-existing geological weaknesses, can trigger large-scale slope failures. Subsequent analyses have shown that the slope movements were detectable through long-term monitoring, but the significance of these precursory signals was underestimated at the time due to a limited understanding of reservoir-triggered mass movements [23]. This underscores the importance of integrating geotechnical monitoring, historical geomorphological analysis, and conservative reservoir management in regions prone to slope instability.

The environmental consequences of catastrophic dam-related events extend beyond immediate loss of life. The sudden release of water and debris can cause long-term alterations to river morphology, soil composition, and sediment distribution. Downstream ecosystems may experience severe habitat degradation, and contaminated materials mobilized by the event can pose additional risks to human health and environmental quality [24]. While infrequent, such catastrophic scenarios highlight the need for robust safety protocols, real-time monitoring technologies, and comprehensive emergency response planning.

Climate change is increasingly recognized as a significant factor influencing the long-term reliability, safety, and environmental performance of hydropower systems. Variations in precipitation regimes, intensified droughts, altered snowfall patterns, and shifting seasonal runoff dynamics directly affect hydropower production capacity and reservoir management strategies [25]. In mountain regions, declining snowpack and earlier spring melt reduce the temporal storage that historically supported stable hydropower generation. Conversely, increases in extreme rainfall events can lead to rapid inflow surges, challenging reservoir operators and elevating flood risks [26].

Projected increases in hydrological variability impose structural and operational stresses on dams and associated infrastructure. More frequent and intense floods increase spillway usage, intensify erosive forces, and increase sediment inflows, with potential consequences for reservoir siltation, turbine abrasion, and downstream ecosystem disturbance. In regions experiencing prolonged drought, low reservoir levels limit generation capacity while simultaneously exposing structural components, potentially affecting dam stability and hydropower economics [27]. These climate-driven dynamics demand adaptive reservoir management practices, including flexible operating rules, real-time hydrological forecasting, and incorporation of climate projections into safety assessments.

Furthermore, climate change may exacerbate geotechnical hazards that affect dam safety. Increased precipitation intensity can enhance landslide activity on reservoir slopes, while repeated cycles of drought and heavy rainfall may weaken geological formations, increasing the likelihood of mass movements similar to those implicated in historical dam-related disasters. As a result, climate-informed risk assessment frameworks are essential to ensure the resilience of hydropower infrastructure over the coming decades.

Several historical dam failures illustrate the severe consequences of mismanagement of structural, geological, and hydrological factors in hydropower systems. The 1959 Malpasset Dam failure in France resulted from a combination of geological instability and extreme rainfall. The collapse triggered a destructive flood wave that caused more than 400 fatalities, highlighting the importance of detailed geotechnical site characterization and conservative flood design criteria [28]. The event remains a benchmark case in dam engineering regarding the interplay between foundation geology and hydrological loading.

The 1975 Banqiao failure in China, one of the deadliest dam disasters ever recorded, occurred during an extreme rainfall event exceeding design specifications. The collapse of multiple dams in the system produced catastrophic downstream flooding, leading to tens of thousands of deaths and widespread environmental devastation [29]. This tragedy underscores the systemic risks posed by cascade dam networks when exposed to unprecedented climatic extremes.

In 1976, the Teton Dam in the United States failed during its initial filling, releasing the impounded reservoir and causing extensive downstream damage. Investigations attributed the collapse to foundation permeability, inadequate seepage control, and structural deficiencies [30]. The Teton case demonstrates the critical importance of construction quality control, foundation treatment, and staged reservoir filling procedures, all of which remain central to hydropower safety practices today.

2.3 Difficulties in Increasing Renewable Energy Production

Integrating large shares of renewable generation into national grids faces substantial technical, economic, and environmental challenges [3]. Variable and intermittent resources, such as wind and solar, require enhanced

flexibility in both generation and demand management. Although energy storage technologies, demand response programs, and advanced forecasting partially mitigate these issues, scalability is often limited by cost, spatial constraints, and technological maturity [4]. Existing grid infrastructure constitutes a further limitation. Many national transmission and distribution networks were designed for centralized, dispatchable generation and may be insufficiently robust to accommodate high penetration of decentralized renewable sources. Upgrading or expanding grid capacity involves significant CAPEX, regulatory approvals, and extended construction lead times, which can delay the rapid expansion of renewables [5]. Grid stability issues, such as congestion, voltage fluctuations, and reactive power management, also become more pronounced with higher renewable penetration. Economic and policy factors additionally constrain renewable deployment. High upfront investment costs, exposure to volatile material and equipment prices, and uncertainties in electricity market revenues can reduce the financial attractiveness of projects [6]. Government incentives, feed-in tariffs, and long-term power purchase agreements often determine the feasibility of large-scale installations, particularly for nascent technologies. Conversely, the reduction of subsidies or policy instability can slow or limit the growth of renewable energy. Environmental and social considerations also impose practical constraints. The large-scale deployment of renewables may generate land-use conflicts, ecological impacts, and community opposition, particularly for wind and hydroelectric projects. Site-specific factors, including solar irradiance, wind patterns, water availability, and ecological sensitivity, determine the technical and economic feasibility of each project [7]. These spatial and environmental limitations restrict the potential expansion of renewable capacity. In conclusion, while renewable energy is essential for decarbonizing electricity systems, technical, economic, environmental, and social challenges collectively limit the possibility of rapid expansion. Given these constraints, it is difficult to predict that in Italy it will be possible to substantially increase the share of electricity produced from renewable sources.

2.4 Gas-Fired Thermal Capacity Required for Nighttime-Only Charging

The Italian nighttime charging demand is approximately 60 GW. Ensuring security of supply requires sufficient dispatchable thermal capacity capable of rapid response and high availability. Gas-fired power plants remain the dominant technology for this purpose due to their flexibility and relatively low capital costs compared to other dispatchable options [31]. The total investment cost is estimated as a function of installed capacity and specific CAPEX per unit of power.

2.5 Mixed Configuration: Combined-Cycle Gas Turbines and Open-Cycle Gas Turbines

In a mixed configuration, peak demand is supplied by a combination of combined-cycle gas turbines (CCGT) and open-cycle gas turbines (OCGT). CCGT units provide higher efficiency and lower fuel costs during extended operation, while OCGT units are primarily dedicated to short-duration peak events and reserve provision [3]. Assuming a total installed capacity equal to peak demand, the installed capacity is set at 60,000 MW, with an average investment cost of €3.0 million per MW. The resulting total CAPEX in EUR is therefore:

$$C_{\text{tot}}^{\text{mix}} = 60,000 \times 3.0 \times 10^6 = 180 \text{ billion EUR} \quad (1)$$

Converting to USD at an exchange rate of 1 EUR = 1.08 USD, the total investment becomes approximately:

$$C_{\text{tot}}^{\text{mix}} \approx 194.4 \text{ billion USD} \quad (2)$$

This estimate includes engineering, procurement, and construction costs, but excludes fuel costs, operation and maintenance expenses, fuel price variability, and grid connection costs.

3 Cost Analysis of Transmission Infrastructure in Italy

The techno-economic assessment of the Italian transmission network was conducted by evaluating both the line infrastructure (see Table 1) and the substations at different voltage levels. Transmission lines were categorized by voltage and type, including overhead, buried, and HVDC buried lines. Substations were classified as either Air Insulated Switchgear (AIS) or Gas Insulated Switchgear (GIS). Table 2 provides a comprehensive overview of the estimated costs of transmission lines and substations in MUSD, with substation costs calculated based on typical installed capacity per voltage level [32, 33]. Table 3 summarizes the corresponding total investments by voltage level, highlighting the cumulative contribution of transmission lines and substations. As shown, the total investment increases with voltage, reflecting both the higher cost per kilometer of high-voltage lines and the larger substation capacities required for 380–400 kV networks.

3.1 Methodology for Cost Estimation

The cost estimation of the Italian transmission infrastructure considers both the line network and substations across the 150 kV, 220 kV, and 380–400 kV voltage levels. Transmission line costs are calculated on a per-kilometer

basis and vary according to the line type. Standard overhead lines represent typical single-circuit aerial lines, while buried lines are assumed to cost approximately three times the corresponding overhead line. For extra-high voltage networks, HVDC buried lines are also considered, with costs assumed to be approximately four times those of overhead lines. Line lengths are based on Terna reports and other sector literature [32]. Substation costs are expressed in MUSD per MW. Typical installed capacities are assumed to be 100 MW for 150 kV substations, 200 MW for 220 kV substations, and 500 MW for 380–400 kV substations. Both AIS and GIS technologies are included, with costs derived from published sources [33] and converted from Euros to USD using a factor of 1.08.

Table 1. Estimated cost of transmission lines in Italy by voltage level and type (MUSD)

Voltage (kV)	Length (km)	Cost per km (MUSD)	Total cost (MUSD)	Type	Notes
380	11211	11.9	133.5	Overhead	Standard overhead single-circuit line
380	11211	35.7	400.0	Buried	Buried cable, about three times the cost of overhead
380	11211	50.0	560.6	HVDC buried	High-voltage DC buried cable, high reliability
220	11043	4.3	47.5	Overhead	Standard overhead single-circuit line
220	11043	12.9	142.3	Buried	Buried cable, about three times the cost of overhead
150	40000	4.3	172.0	Overhead	Estimated length for lower voltage network
150	40000	12.9	516.0	Buried	Buried cable, higher cost

Note: MUSD = million USD.

Table 2. Estimated total costs of transmission lines and substations in Italy (MUSD)

Category	Voltage (kV)	Asset/Type	Unit Cost	Total (MUSD)	Notes
Transmission lines	380	Line, Overhead	11.9/km	133.5	Standard overhead single-circuit line
	380	Line, Buried	35.7/km	400.0	Buried cable, about 3x overhead cost
	380	Line, HVDC buried	50.0/km	560.6	High-voltage DC buried cable, high reliability
	220	Line, Overhead	4.3/km	47.5	Standard overhead single-circuit line
	220	Line, Buried	12.9/km	142.3	Buried cable, about 3x overhead cost
	150	Line, Overhead	4.3/km	172.0	Estimated length for lower voltage network
	150	Line, Buried	12.9/km	516.0	Buried cable, higher cost
Substations	150	Substation, AIS	0.81/MW	81.0	100 MW typical substation, AIS
	150	Substation, GIS	1.24/MW	124.0	100 MW typical substation, GIS
	220	Substation, AIS	0.97/MW	194.0	200 MW typical substation, AIS
	220	Substation, GIS	1.51/MW	302.0	200 MW typical substation, GIS
	400	Substation, AIS	1.24/MW	620.0	500 MW typical substation, AIS
	400	Substation, GIS	1.73/MW	865.0	500 MW typical substation, GIS

Note: MUSD = million USD; AIS = Air Insulated Switchgear; GIS = Gas Insulated Switchgear.

Table 3. Estimated total investment for transmission lines and substations in Italy by voltage level (MUSD)

Voltage (kV)	Components Included	Total (MUSD)	Notes
150	Lines (Overhead + Buried) + Substations (AIS + GIS)	893.0	Lines: 172 + 516; Substations: 81 + 124
220	Lines (Overhead + Buried) + Substations (AIS + GIS)	685.8	Lines: 47.5 + 142.3; Substations: 194 + 302
380–400	Lines (Overhead + Buried + HVDC buried) + Substations (AIS + GIS)	2579.1	Lines: 133.5 + 400 + 560.6; Substations: 620 + 865

Note: MUSD = million USD; AIS = Air Insulated Switchgear; GIS = Gas Insulated Switchgear.

3.1.1 Detailed cost distribution methodology

Total line costs are obtained by summing all line types for each voltage level, while substation costs are calculated by multiplying the unit cost per MW by the assumed substation capacity. The overall investment per voltage level is then determined by adding the total costs of the lines and substations. Table 2 presents the estimated costs for transmission lines and substations separately. Based on the assumptions described above, the total estimated investments for each voltage level are 893 MUSD for 150 kV networks (Lines: 688 MUSD, Substations: 205 MUSD), 685.8 MUSD for 220 kV networks (Lines: 189.8 MUSD, Substations: 496 MUSD), and 2579.1 MUSD for 380–400 kV networks (Lines: 1094.1 MUSD, Substations: 1485 MUSD), as summarized in Table 3. Figure 1 illustrates the detailed distribution of costs by component for each voltage level, highlighting the relative contributions of overhead, buried, and HVDC lines, as well as AIS and GIS substations. These results indicate that investment in extra-high voltage networks is primarily driven by substation costs, particularly the cost of GIS technology, while lower voltage networks are predominantly influenced by line construction costs. The analysis provides a quantitative foundation for planning and prioritizing investments in the Italian transmission network, supporting cost-benefit evaluations for network expansion and modernization.

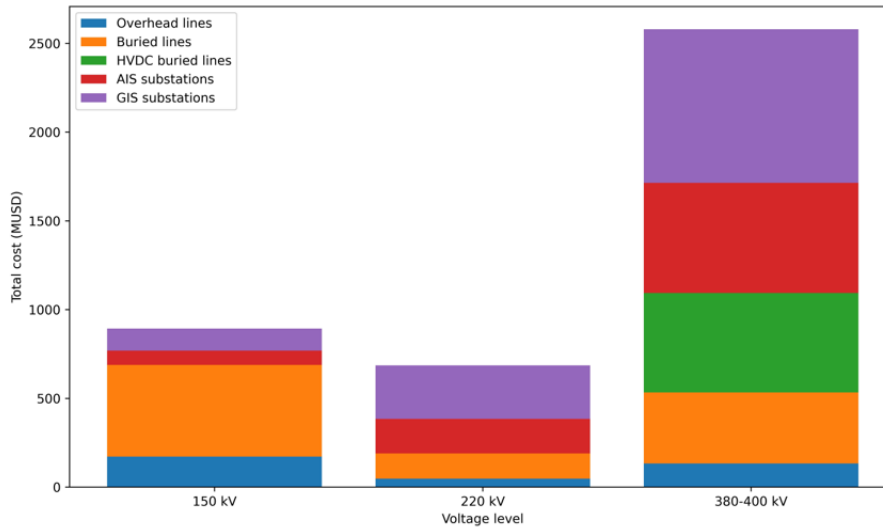


Figure 1. Detailed cost distribution of transmission infrastructure in Italy by voltage level, including overhead, buried, and HVDC transmission lines, as well as Air Insulated Switchgear (AIS) and Gas Insulated Switchgear (GIS) substations

3.2 Cost Estimation of the 380 V Distribution Network

The 380 V distribution network, representing the final low voltage level for residential and small industrial consumers in Italy, was analyzed to estimate the investment required for both distribution lines and local substations or transformer cabins. Most distribution lines are assumed to be underground, reflecting modern urban infrastructure, while aerial lines are considered negligible at this voltage level. Based on sector data, the average cost of underground lines is approximately 50,000 USD per kilometer, with a total estimated network length of 150,000 km. This results in a total line investment of 7,500 MUSD. Local substations, typically consisting of small transformers with capacities ranging from 0.5 to 1 MVA, are assumed to number around 12,000 units. The average cost per substation is estimated

at 0.1 MUSD, yielding a total investment of \$1.2 billion for the substations. Unlike higher voltage networks, AIS and GIS technologies are not applicable at this level due to the small scale and low voltage of these installations. Combining the line and substation costs, the overall investment required for the 380 V distribution network is approximately 8,700 MUSD. Specifically, underground lines account for 7,500 MUSD (86% of the total), while transformer cabins account for 1,200 MUSD (14%). Table 4 summarizes the cost distribution, including the totals, providing a clear view of the investment allocation for low-voltage distribution infrastructure in Italy.

Table 4. Estimated costs of the 380 V distribution network in Italy (MUSD), including totals

Category	Voltage (V)	Asset	Type	Total Cost (MUSD)	Notes
Distribution Lines	380	Line	Underground	7,500	Underground network, average cost \$50,000/km, estimated length 150,000 km
Substations/Transformers	380	Transformer/Cabin	Standard	1,200	Neighborhood substations, typical capacity 0.5–1 MVA, 12,000 units
Total	-	-	-	8,700	Sum of lines and transformers/substations

Note: MUSD = million USD.

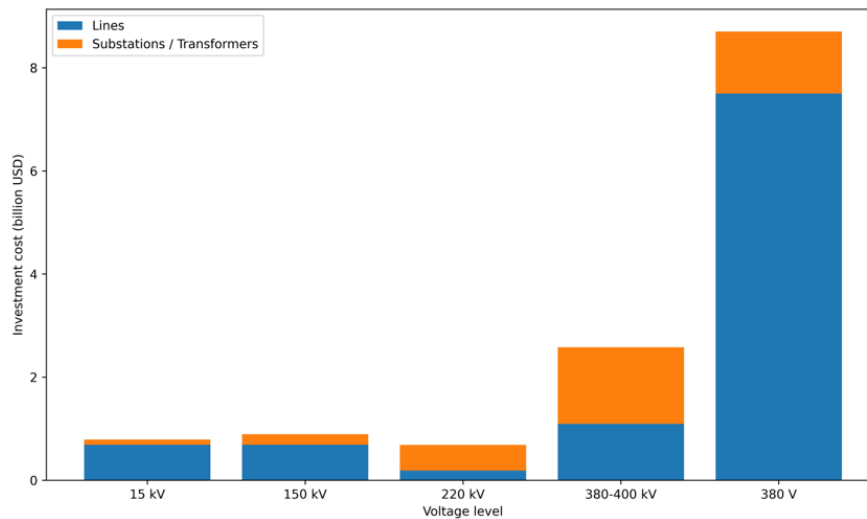


Figure 2. Overall investment costs of the Italian electricity network by voltage level, showing the contribution of line infrastructure and substations or transformers

3.3 Overall Cost Assessment of Italian Transmission and Distribution Networks

An integrated analysis of the Italian electricity network was conducted to estimate the total investment required across all voltage levels, ranging from high-voltage transmission to low-voltage distribution. The assessment considers both line infrastructure and substations or transformer cabins, expressed in USD, and provides a quantitative basis for network planning and modernization. For high- and extra-high-voltage networks (150–400 kV), line costs were estimated based on type (overhead, buried, or HVDC buried) and typical lengths reported in the literature [32]. Substation costs were expressed per installed MW, with AIS and GIS technologies included. The total estimated investments for these voltage levels are approximately \$0.893 billion for 150 kV (lines: \$0.688 billion, substations: \$0.205 billion), \$0.686 billion for 220 kV (lines: \$0.190 billion, substations: \$0.496 billion), and \$2.579 billion for 380–400 kV (lines: \$1.094 billion, substations: \$1.485 billion). For low voltage networks at 15 kV (typical urban distribution), line costs were dominated by overhead and buried cables, while substation costs were based on 50 MW AIS and GIS units. The total estimated investment is approximately \$0.791 billion (lines: \$0.688 billion, substations: \$0.103 billion). For the 380 V distribution network, which serves residential and small industrial customers, underground lines account for the majority of the cost, with small transformer cabins representing a smaller portion. The total estimated investment is approximately \$8.7 billion (lines: \$7.5 billion, transformers: \$1.2 billion). Combining all voltage levels, the overall investment required for the Italian electricity network is approximately \$13.648 billion, with line infrastructure accounting for \$10.660 billion (78% of the total) and substations or transformers representing \$2.988 billion (22%). Expressed on a larger scale, this corresponds to approximately \$0.0136 trillion, underscoring the magnitude of investment required for the entire national network.

These estimates can be compared with the physical characteristics of the Italian transmission system. According to Terna, the network encompasses over 45,000 km of primary transmission lines, including 380 kV and other high voltage lines, and approximately 49,000 km when smaller lines are included. Strategic projects, such as the Italy-France interconnection, involve the addition of underground high-voltage segments. The investment figures obtained in this study are consistent with the scale of the network, indicating that most costs are associated with long-distance, high-voltage lines and low-voltage urban distribution, while substations and local transformers, although essential, account for a smaller fraction of total expenditure. These results emphasize the importance of maintaining and upgrading the Italian network to ensure reliable operation and integration with European energy markets. Figure 2 provides a visual comparison of cost contributions by voltage level and asset type, highlighting the dominant role of low-voltage distribution lines in the overall investment.

3.4 Total Cost Estimate for Italian Grid Reinforcement under Nighttime-Only Charging

This section summarizes the total investment required to support nighttime-only charging of a fully electric vehicle fleet in Italy. The investment associated with additional power generation capacity, based on a mixed CCGT and OCGT configuration, was estimated at approximately \$194.4 billion, or \$0.194 trillion. The reinforcement of the Italian transmission and distribution network across all voltage levels was estimated at approximately \$13.648 billion, corresponding to about \$0.0136 Tera. Summing up these two contributions, the total investment required for night-time-only charging amounts to approximately \$208.0 billion, that is about \$0.208 Tera. The overall cost is clearly dominated by the installation of additional power generation capacity, while grid reinforcement represents a smaller but still non-negligible fraction of the total expenditure.

4 Daytime-Only Fast Charging for a Fully Electric Vehicle Fleet

Based on previous studies, the power demand associated with daytime-only fast charging of a fully electric vehicle fleet is estimated to be approximately four to five times the 2024 network peak, corresponding to a range of 240,000 to 300,000 MW. In this scenario, the implementation of low- and medium-voltage networks (380 V to 220 kV) is avoided, as fast charging stations are designed to handle very high-power levels. Consequently, the investment costs associated with the construction and upgrade of traditional distribution networks are not included in this analysis. It should also be noted that the costs of the fast-charging stations themselves are excluded from the assessment, as the focus is on the impact of high-power demand on generation and transmission infrastructure.

4.1 Cost Assessment of Italian Transmission and Low Voltage Networks (Excluding 380 V Distribution)

This section presents a comprehensive assessment of the investment required for the Italian transmission and low voltage networks, excluding the final 380 V distribution network. The analysis covers high-voltage (150–400 kV) and low-voltage (15 kV) networks, considering both line infrastructure and substations, with costs expressed in billions of USD. For high- and extra-high-voltage networks, line costs were estimated based on type (overhead, buried, or HVDC buried) and typical lengths reported in the literature [32]. Substation costs were expressed per installed MW, with both AIS and GIS technologies included. The total estimated investments are approximately \$0.893 billion for 150 kV (lines: \$0.688 billion, substations: \$0.205 billion), \$0.686 billion for 220 kV (lines: \$0.190 billion, substations: \$0.496 billion), and \$2.579 billion for 380–400 kV (lines: \$1.094 billion, substations: \$1.485 billion). For low-voltage networks at 15 kV, which typically serve urban and semi-urban areas, the investment is dominated by line infrastructure, including both overhead and buried cables, while substations are based on 50 MW AIS and GIS units. The total estimated investment is \$0.791 billion (lines: \$0.688 billion, substations: \$0.103 billion). Combining all voltage levels up to 15 kV and including high-voltage transmission up to 400 kV, the total estimated investment required for the Italian electricity network (excluding 380 V distribution) is approximately \$4.949 billion. Line infrastructure represents \$2.660 billion (54% of the total), while substations account for \$2.289 billion (46%). Expressed on a larger scale, this corresponds to approximately \$0.00495 trillion. These estimates are consistent with the physical characteristics of the Italian transmission system. According to Terna, the primary transmission network exceeds 45,000 km for high voltage lines and approaches 49,000 km when smaller lines are included [32]. The results indicate that most costs are associated with high-voltage lines and substations, while low-voltage infrastructure contributes a smaller, yet significant, portion of the overall investment. This highlights the importance of ongoing network maintenance and modernization to ensure reliability and integration with European energy markets. Figure 3 provides a visual comparison of cost contributions by voltage level and asset type, illustrating the relative contributions of line infrastructure and substations across high and medium voltage networks.

4.2 Open-Cycle Gas Turbines-Only Configuration (Minimum Capital Expenditure Case)

An alternative approach consists of supplying peak demand exclusively with OCGT. OCGT units are characterized by lower capital costs, short construction times, and high ramping capability, making them suitable for peak-load and capacity adequacy purposes [34–36]. For the installed capacity of 60,000 MW, the specific investment cost

is 2.0 MUSD/MW, which corresponds to approximately 2.16 MUSD/MW at an exchange rate of 1 EUR = 1.08 USD. The total investment cost for this configuration is therefore 120 billion EUR, or approximately \$129.6 billion. This configuration yields a minimum CAPEX, but the lower efficiency of OCGT units implies higher operational and fuel costs, as well as increased carbon emissions during operation, compared to a mixed or CCGT-dominated configuration.

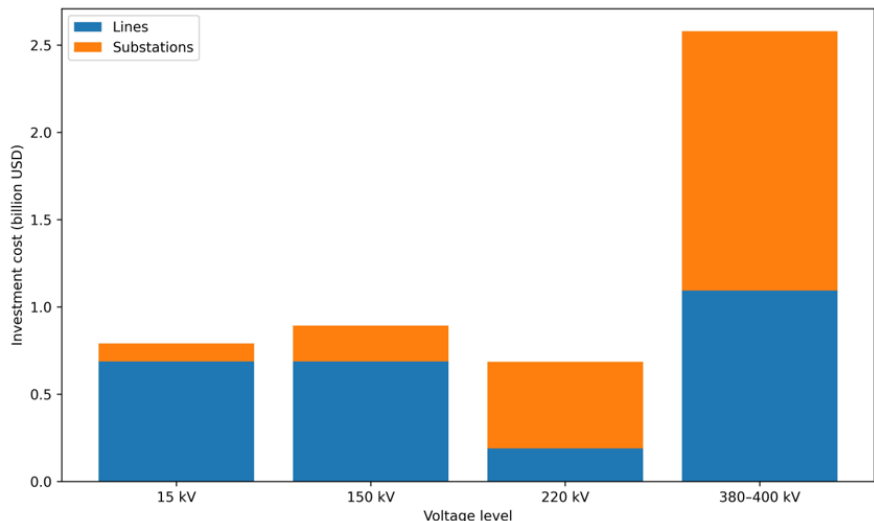


Figure 3. Investment costs of the Italian electricity network by voltage level, excluding the 380 V distribution network, showing the contribution of line infrastructure and substations

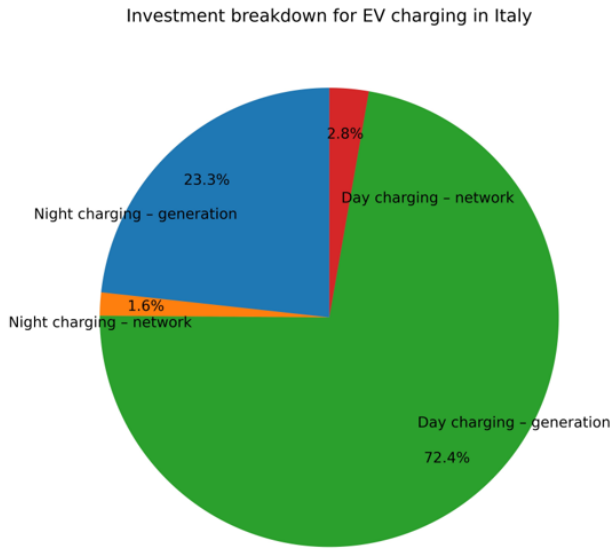


Figure 4. Overall investment breakdown for nighttime and daytime electric vehicle charging in Italy, showing the relative contributions of generation capacity and network reinforcement

4.3 Cost Required for Daytime-Only Fast Charging

Based on the previously calculated values for a 60,000 MW installed capacity, the total investment required to supply daytime-only fast charging for a fully electric vehicle fleet can be estimated proportionally for a higher power demand of 280,000 MW. Scaling the OCGT-only configuration, the total investment for generation is: \$129.6 billion $\times$ (280,000/60,000) $\approx$ \$604.8 billion, or approximately \$0.605 Tera. The investment required for the Italian transmission and low-voltage networks, up to 15 kV, excluding 380 V distribution, was previously estimated at \$4.949 billion for 60,000 MW. Scaling proportionally to 280,000 MW, the total network investment becomes: \$4.949 billion $\times$ (280,000/60,000) $\approx$ \$23.1 billion, or approximately \$0.0231 Tera. Combining these two components, the overall

investment required to support daytime-only fast charging of 280,000 MW is therefore approximately \$627.9 billion, or about \$0.628 Tera. This estimate focuses exclusively on the additional generation and transmission capacity required to meet the high-power demand of fast charging, excluding the costs of fast charging stations and 380 V distribution upgrades. The results indicate that most of the investment is associated with generation capacity, while network reinforcement contributes a smaller, yet non-negligible, portion of the total expenditure.

5 Total Investment for Nighttime and Daytime Charging

This section summarizes the total investment required to support nighttime-only and daytime-only charging of a fully electric vehicle fleet in Italy. For nighttime-only charging, the total investment is approximately \$208.0 billion (\$0.208 Tera), including \$194.4 billion (\$0.194 Tera) for generation and \$13.6 billion (\$0.014 Tera) for network reinforcement. For daytime-only fast charging at 280,000 MW, the total investment is roughly \$627.9 billion (\$0.628 Tera), with \$604.8 billion (\$0.605 Tera) for generation and \$23.1 billion (\$0.023 Tera) for network reinforcement. Summing both scenarios, the combined investment required for nighttime and daytime charging amounts to approximately \$835.9 billion, or about \$0.836 Tera. This comparison highlights the dominant role of generation capacity in overall expenditure, with network reinforcement contributing a smaller, yet still significant, fraction. Figure 4 shows the investment breakdown between generation capacity and network reinforcement for nighttime and daytime electric vehicle charging in Italy, based on the assumptions and cost data reported in [3, 32, 33].

6 Discussion

6.1 Energy Demand Associated with Full Electrification

The analysis presented in this work indicates that the amount of electrical energy required to convert the entire Italian vehicle fleet from fossil-fuel and hybrid propulsion systems to a fully electric configuration would be extremely large. This conclusion applies both to scenarios in which vehicle charging occurs exclusively during nighttime hours and, even more critically, to scenarios that include both nighttime and daytime charging.

In particular, daytime charging would generate very high peak power demand. Such peaks would place significant stress on the national electrical grid and would require substantial expansion of generation capacity, transmission infrastructure, and distribution networks.

6.2 Implications for the Electrical Distribution Network

A large-scale transition to electric vehicles would also have significant implications for the residential electrical distribution system. The number of supply points connected to the electrical network would likely need to increase substantially. In practical terms, the number of users connected to the grid could increase by approximately 40-50%.

Moreover, each new residential user would require a higher installed power rating compared to the traditional 2.5 kW household supply commonly used in Italy. To support electric vehicle charging, installed capacities of at least 5 kW would likely be necessary.

Therefore, the total installed capacity of the national electrical system would need to increase dramatically. The calculations presented in this work suggest that the maximum installed power could exceed the maximum power currently deliverable by the Italian electrical grid by up to five times.

6.3 Economic Implications of Grid Expansion

The expansion of the electrical infrastructure required to support such a transition would involve extremely high costs. These costs would primarily be associated with upgrades to generation capacity, transmission lines, and local distribution networks.

In practice, the financial burden associated with these investments would largely fall on taxpayers or electricity consumers through higher tariffs. The scale of these required investments raises questions about the economic feasibility of a rapid transition to a fully electric vehicle fleet within the current structure of the national economy.

Based on the calculations presented in this study, the scale of the required investment appears significantly larger than what the Italian economic system could reasonably sustain.

6.4 Industrial and Regulatory Context

The transition toward electric mobility must also be considered within the broader context of recent developments affecting the national automotive industry. Historically, the automotive sector represented one of the most important pillars of Italian manufacturing.

Over the past decades, however, several policy measures have contributed to structural changes in this sector. These include high fuel taxes, progressively stricter emissions regulations, and policies that have strongly penalized diesel engines despite their demonstrated contribution to reducing particulate emissions in certain regions, including the Po Valley.

In addition, the introduction of new safety and monitoring regulations has required the widespread adoption of driver-assistance and driver-monitoring technologies. In some cases, these regulatory measures have been implemented without sufficient long-term testing or comprehensive evaluation of their broader implications.

These developments have shortened vehicle replacement cycles, encouraging consumers to purchase new vehicles earlier than would otherwise be necessary. In the case of emissions-related policies, some regulatory changes may also have indirectly increased fuel consumption in certain operating conditions.

6.5 Technological and Safety Considerations

Recent trends in vehicle design have also introduced new technological elements that raise safety and usability concerns. Examples include the widespread use of large interactive touchscreens for essential vehicle functions, the adoption of retractable electronic door handles that may prevent exit in the absence of electrical power, and the deployment of certain safety devices that may not have undergone sufficient long-term validation.

Although many of these technologies are intended to improve safety or user experience, their real-world implications require careful evaluation to ensure that they do not introduce unintended risks.

6.6 Limitations and Perspectives for Future Research

The author acknowledges the possibility that the estimates presented in this work may represent a pessimistic scenario. It is possible that some assumptions used in the calculations may overestimate the actual energy requirements associated with a fully electric vehicle fleet.

For this reason, the results presented here should be interpreted primarily as a starting point for further analysis. Additional studies, incorporating updated technological developments, refined energy models, and alternative infrastructure strategies, will be necessary to fully evaluate the feasibility and long-term implications of large-scale electrification of the national vehicle fleet.

7 Conclusions

7.1 Summary of Main Findings

The analysis presented in this work highlights the substantial technical and economic challenges associated with fully converting the Italian vehicle fleet to electric power. The results indicate that the required increase in both installed power capacity and the number of supply points would be unprecedented in scale, even under conservative assumptions.

Nighttime-only charging would already require a doubling of the current electrical grid in terms of user connections. If both nighttime and daytime charging were considered, peak power demand could reach up to five times current levels. Such requirements appear to exceed the realistic expansion capabilities of the existing infrastructure within the foreseeable future.

7.2 Economic Implications

From an economic perspective, the investment required to redesign the national electrical grid to accommodate a fully electric vehicle fleet would be extremely large. The financial burden associated with these infrastructure upgrades would likely fall largely on taxpayers or electricity consumers and would require long-term financial commitments.

The scale of the required investments suggests that a rapid, uniform transition to fully electric mobility may not be compatible with the current structure and capacity of the Italian energy system.

7.3 Policy Implications

These findings highlight the importance of grounding energy and mobility policies in quantitative assessments rather than relying on simplified or overly optimistic assumptions. The results suggest that current policy orientations toward a rapid and mandatory transition to electric vehicles may not fully account for the broader technical and economic constraints of the national electrical grid.

While electric mobility can contribute to emissions reduction in certain contexts, the large-scale impacts on infrastructure requirements, electricity generation capacity, and grid stability must be carefully evaluated. Without such considerations, the transition could generate unintended consequences that offset part of the expected environmental benefits.

7.4 Alternative Transition Pathways

The analysis also suggests that a diversified technological strategy may be more feasible than a transition based on a single propulsion technology. Hybrid powertrains, synthetic fuels, hydrogen-based solutions, and further improvements in internal combustion engines could potentially play complementary roles during the transition phase.

Such a diversified approach could reduce pressure on the electrical grid while enabling a more gradual, economically sustainable evolution of the vehicle fleet. In addition, a mixed technological strategy may help distribute the economic burden more evenly across consumers, industry, and public institutions.

7.5 Limitations and Future Research

The work presented here should be considered an initial and deliberately conservative assessment. More refined models, improved datasets, and updated cost estimates will be necessary to further increase the accuracy of future analyses.

Nevertheless, the results indicate that the scale of the transformation required for full electrification of the vehicle fleet may currently be underestimated in public debate. Future research should therefore incorporate not only energy demand and infrastructure constraints but also broader economic, industrial, and social implications. Such integrated analyses will be essential to support transition strategies that are technically feasible, economically sustainable, and consistent with the long-term development of the national industrial system.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

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